CLOCK IN: THE 3RD SOLANA MOBILE HACKATHON

HAPTIX

The Biometric Hardware Enclave Smart Wallet for Solana Seeker & Saga

Seedless Passkeys (Secp256r1) | 2-of-2 Hardware Quorum | Autonomous AI Agent Sessions | Seeker SKR Token Ready | Gasless Devnet Paymaster

01 // PROBLEM STATEMENT

Core Market Pain

Mobile Web3 Is Trapped Between Friction and Vulnerability

Seed Phrase Liability

12–24 words on paper remain the #1 churn point in Web3 onboarding. A single backup breach results in 100% loss of funds, with zero recovery path for everyday users.

Security & Prompt Fatigue

Current mobile wallets treat a $0.05 micro-swap with the same friction as a $25,000 liquidation. Constant biometric popups lead to blind-signing malicious transactions.

The Mobile AI Agent Dilemma

Autonomous trading agents on Solana currently require either continuous human biometric approval or unconstrained hot keys stored in device memory vulnerable to drainers.

02 // THE SOLUTION

4-Layer Smart Wallet Engine Enforced on Solana

Anchor Bytecode Policy

1. Fast Lane (≤ 0.1 SOL)

Micro-transactions authorize via 1-tap biometric passkey directly through the phone's Secure Enclave. Confirms on Solana Devnet in ~400ms with zero seed phrases.

2. 2-of-2 Hardware Quorum (> 0.1 SOL)

High-value outflows mathematically demand two independent hardware signatures (Phone + Mac Touch ID / YubiKey) verified via native Secp256r1 precompile.

3. AI Agent Delegation

Delegates bounded session keys to autonomous trading bots (Jupiter DCA) with on-chain spending caps and time expiries. Zero user interruptions required.

Cryptographic Feats & Mobile Primitives

Zero Compromises

Native Secp256r1 Precompile Introspection

Instead of costly custom curve math, Haptix utilizes Solana's native `Secp256r1SigVerify1111111111111111111111111` precompile instruction. The program introspects the instructions sysvar to verify the signature header and normalize low-S (`s ≤ n/2`).

Dual-Device Vault Mode via Out-of-Band Relay

High-value actions broadcast an ephemeral verification request over an encrypted local/remote WebRTC relay to a Mac Secure Enclave (`vault.html`). Neither key ever shares memory space.

Zero-SOL Gasless Onboarding

Our embedded Paymaster sponsors account creation rent and immediately drips a 0.10 SOL starter test balance so users never encounter a dead-end `$0.00` screen.

Timelocked Guardian Recovery & Panic Freeze

Emergency circuit breaker permits designated social guardians to trigger an on-chain `panic_freeze`, while lost devices can be recovered through a 24-hour timelocked passkey rotation.

04 // ECOSYSTEM ALIGNMENT

18 Passing Tests

Seeker (SKR) Token Architecture Ready ($10,000 Bonus Track)

Risk-Adaptive SKR Policy

- ≤ 50 SKR: 1-tap instant biometric passkey.
- > 50 SKR: Automatically escalates to 2-of-2 hardware quorum.

Tested extensively in our automated test harness with full mock mints.

Genesis Reward Distribution

Onboarding automatically initializes a 100 SKR Genesis Reward balance, demonstrating seamless SPL token reception and multi-asset threshold management.

Solana Mobile Stack (SMS) Ready

Architected for direct compatibility with the Seeker Seed Vault, Android BiometricPrompt hardware keystores, and Solana dApp Store publishing guidelines.

05 // SYSTEM TOPOLOGY

End-to-End System Architecture

React Native + Anchor Rust

Mobile Client (React Native / Expo 55)

- Native Android BiometricPrompt & WebAuthn / P-256 Signer
- Account Switcher (Personal Everyday vs Multi-Device Vault)
- Rare-UI StepPlayer Onboarding & Rolling Odometer Counters
- Local Treasury Paymaster & Out-of-Band Relay Client
- Pitch-black Cyber-minimalist Design System (0 Emojis, Clean Haptics)

On-Chain Smart Program (Anchor 0.31)

- Program ID: `9wC1ok8d4PrBDG3TS7iyEu9QxMCYv6BKrdLb3q4yZbke`
- Counterfactual Off-Curve PDA Derivation
- Sysvar Instructions Introspection for Precompile Verification
- Dynamic Spend Threshold Evaluation (1-Tap vs 2-of-2 Quorum)
- Bounded Agent Allowance & 24h Timelocked Guardian Recovery

06 // COMPETITIVE LANDSCAPE

Why Haptix Leads the Next Generation

Feature Matrix

Capability	Phantom / Solflare	Backpack	Safe (EVM)	Haptix (Solana Mobile)
Hardware Passkeys (P-256)	No (Seed Phrase)	No (Seed Phrase)	WebAuthn (High Gas)	Native Secp256r1 Precompile
Risk-Adaptive Spend Policy	No (Single Key)	No	Complex Multisig	1-Key vs 2-of-2 Quorum
Out-of-Band Hardware Pair	No	No	No	Phone + Mac Touch ID / YubiKey
AI Agent Session Delegation	No	No	Session Keys	Bounded On-Chain Allowance
Gasless Starter Balance	No ($0 Dead-End)	No	No	0.1 SOL Auto-Funder
Seeker SKR Token Native Support	Basic SPL	Basic SPL	None	Architecture & Thresholds Ready
Emergency Panic Freeze	No	No	Guard Module	Instant On-Chain Freeze

Proven Readiness

Targeting 150,000+ Solana Seeker Pre-Orders

150,000+ Seeker Pre-Orders

The largest pre-ordered crypto hardware distribution in history across 50+ countries. Every device ships with an Android hardware enclave ready for Haptix.

100% Passing Tests on Devnet

57+ comprehensive on-chain integration tests passing live on Devnet. Anchor program verified and operational under Program ID `9wC1ok8d4P....`

Standalone Android APK Ready

Full 131MB release APK prebuilt and verified. Sideloadable instantly on physical Android & Seeker phones with zero complex build toolchains required.

08 // REPRODUCIBILITY

Judge Runbook

Complete Verification in Under 2 Minutes

1-Click Test Harness Execution

```
$ cd solana-mobile/haptix-mobile-app
$ npm test
```

- 14 Test Suites Executed
- 57 Live Devnet Checks Passed
- 18 Seeker SKR Policy Validations Passed

Direct Artifact Links

- **GitHub Repo**: `github.com/ritesh59697/haptix-mobile-app`
- **Anchor Program**: `9wC1ok8d4PrBDG3TS7iyEu9QxMCYv6BKrdLb3q4yZbke`
- **Solana Explorer**: Verified on Devnet Cluster
- **EAS Build Link**: Verified Expo Cloud Build Receipt

THE FUTURE OF MOBILE SOLANA

Turning Solana Seeker into the Safest Smart Wallet on Earth.

No seed phrases. No blind signing. No unconstrained bots.

github.com/ritesh59697/haptix-mobile-app

explorer.solana.com/address/9wC1ok8d4PrBDG3TS7iyEu9QxMCYv6BKrdLb3q4yZbke